

Project Proposal

Project Title	Analyzing the Relationship Between Air Quality and Behavioral Health Indicators Using Data Visualization
Student Name(s) and Net ID(s)	Vinay Kumar – vk2771 Antar Espadas Rodríguez - ae3144 Dharun Ramesh- dr4075
Course Name and Section	CS-GY 6313 Information Visualization, Section B
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Team Name	Team Visualizers

Analyzing the Relationship Between Air Quality and Behavioral Health Indicators Using Data Visualization

Problem Statement / Background

In the US, the increase in obesity and associated chronic illnesses is a serious public health concern. Environmental factors like air quality also have a big impact on public health outcomes, even if individual behavior like nutrition and physical activity are well-known contributors. Inadequate air quality can deter people from exercising outside, which could increase obesity rates and deteriorate general health.

Using extensive public information, this research seeks to investigate the connection between obesity, physical activity, and air quality. Our objective is to present a data-driven viewpoint on the intersection of behavioral and environmental elements by visualizing demographic variances, regional patterns, and temporal trends. This is important because knowing these trends can help with targeted interventions to improve community health outcomes, public health policy, and urban planning.

Readers should be aware that there are two main components to the context: (1) data on behavioral health obtained from national surveys (BRFSS) and (2) measurements of air quality obtained from environmental monitoring systems. Combining these datasets provides a distinctive perspective for studying the relationship between environmental factors and public health outcomes and behaviors.

Objectives

- Analyze geographic and temporal patterns in air quality and obesity indicators.
- Visualize demographic disparities in behavioral risk factors.
- Explore potential correlations between air pollution levels and physical activity/obesity prevalence.
- Develop an interactive dashboard to enable stakeholders to explore trends across time, location, and demographic groups.

Dataset(s)

1. Nutrition, Physical Activity, and Obesity — BRFSS

- Source: <https://catalog.data.gov/dataset/nutrition-physical-activity-and-obesity-behavioral-risk-factor-surveillance-system>
- Collected by: U.S. Centers for Disease Control and Prevention (CDC)
- Description: Annual Behavioral Risk Factor Surveillance System data capturing self-reported health behaviors and conditions.
- Attributes to Use: YearStart, YearEnd, LocationAbbr, LocationDesc, Class, Topic, Question, Data_Value, Sex, Age, Education, Income, Race/Ethnicity.
- Preprocessing Needed: Filtering relevant questions (e.g., obesity rates, physical activity frequency), handling missing values, normalizing categorical labels, and aggregating data by state and year.

2. Air Quality Data

- Source: <https://catalog.data.gov/dataset/air-quality>
- Collected by: U.S. Environmental Protection Agency (EPA) / NYC Environmental agencies
- Description: Contains annual and seasonal measures of air pollutant concentrations (e.g., PM2.5, ozone) across locations.
- Attributes to Use: Geo Place Name, Time Period, Start_Date, Data Value (pollution index), Indicator ID, Measure.
- Preprocessing Needed: Temporal alignment with BRFSS data, location matching (e.g., by state or region), handling missing readings, and aggregating to consistent time intervals.

Domain Questions

The following questions form a logical progression from descriptive to exploratory analysis, each designed to be answered through data visualization:

1. How have obesity and physical activity rates varied geographically across U.S. states over time?
2. What are the temporal trends in air quality measures (e.g., PM2.5 levels) across different regions?
3. Are there visible correlations between poor air quality and lower physical activity levels across New York?
4. How do obesity rates differ across demographic groups (e.g., income, race, education) within regions of varying air quality?
5. Are there specific time periods or regions where declining air quality coincides with changes in obesity or activity rates?

6. Which geographic regions experience the most significant fluctuations in pollution levels over time, and what local factors contribute to this variation?
7. Can we identify clusters of regions with both high pollution levels and high obesity prevalence using geospatial visualizations?
8. How do changes in air quality over time relate to changes in physical activity and obesity rates, when visualized side by side?
9. How do industrial vs. residential areas differ in pollution exposure and trends across the years?

Proposed Technologies and Programming Languages

- Programming Languages: Python (primary), SQL (for data cleaning if needed), JS
- Visualization Libraries: D3, VegaLite, Geopandas, Plotly, Matplotlib, Seaborn, D3, Vegalite
- Data Processing: Pandas, NumPy
- Dashboard: Plotly Dash or Tableau, Streamlit
- Version Control: Git/GitHub
- Environment: Jupyter Notebook / VS Code